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Roll No

IT-4002-CBGS

B.E. IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Database Management System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is data model? List a few data models that you know.
b) Define Database Management System (DBMS). What are the major component of this system? Explain each component.
2. a) What is Relational Algebra? Enlist and explain the fundamental operations of Relational Algebra.
b) Explain disadvantages of conventional file based system compared to Database Management System.
3. a) Define E-R diagram. Draw an E-R diagram for library management system. Assume relevant entities and attributes for given system.
b) What is aggregate function? Explain any five aggregate functions with example.
4. a) Consider the following relational schema:
Employee (Employee_id, F_name, L_name, Salary, Joining_Date, Department)
Incentive (Employee_ref_id, Incentive_date, Incentive_amount)

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Write the following sql query:

- i) Get all employee details from the employee table order by First_Name Ascending and Salary descending.
 - ii) Get department, total salary with respect to a department from employee table.
 - iii) Delete employee data from employee table who got incentives in incentive table.
 - iv) Get employee detail from employee table whose employee name is "Rakesh".
- b) Explain Query Processing with suitable example.
5. Consider the universal relation $R\{A,B,C,D,E,F,G,H,I\}$ and the set of functional dependencies $F=\{A,B\}\rightarrow\{C\}, \{A\}\rightarrow\{D,E\}, \{B\}\rightarrow\{F\}, \{F\}\rightarrow\{G,H\}, \{D\}\rightarrow\{I,J\}$. What is the key for R? Decompose R into 2NF, then 3NF relations.
6. a) What is normalization? Explain 2nd Normal form with the help of an example.
b) What is BCNF? Explain with examples.
7. a) What is serializability? Explain view and conflict serializability with example.
b) Explain following relational algebra operation:
i) Natural join operation
ii) Selection and projection operation
8. Write short notes:
a) Distributed databases
b) Two phase locking protocol
c) Log based recovery

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